

Satya Himavanth T

Sri Venkataramana Colony, Vanastalipuram, Hyderabad.

9492790628 | himavantht31@gmail.com

 <https://github.com/SatyaHimavanth>

 <https://www.linkedin.com/in/satyahimavanth-tripurari/>

Objective

AI Research Engineer specializing in autonomous and agentic systems, with hands-on experience designing, evaluating, and deploying multi-agent architectures for real-world automation. My work sits at the intersection of LLM-based reasoning, system orchestration, and applied research, with a strong focus on translating experimental ideas into scalable, production-ready systems. Passionate about building intelligent agents that interact with software, data, and users in context-aware ways.

Experience

- **Tata Consultancy Services Limited**

06-02-2025 - Present

AI Engineer / Research Engineer – Autonomous Agents

- Designed and implemented autonomous AI agents capable of task planning, tool usage, and multi-step reasoning using LLMs and agent orchestration frameworks.
- Conducted applied research on multi-agent coordination, supervisor-based workflows, and agent reproducibility, converting findings into deployable platforms.
- Built research PoCs exploring emerging frameworks (LangChain, LangGraph, Google ADK, A2A) to evaluate agent reliability, scalability, and orchestration patterns.
- Developed agents that interact with software systems and APIs, enabling natural-language-driven automation across databases, ticketing systems, and enterprise tools.
- Established evaluation loops for agent performance, focusing on accuracy, task completion, hallucination reduction, and response consistency.
- Collaborated closely with backend engineers, cloud teams, and stakeholders to align research outcomes with product and client requirements.
- Deployed and managed AI systems on GCP (Cloud Run, Cloud SQL, VPCs, Load Balancers) with CI/CD pipelines and containerized workflows.
- Led internal knowledge-sharing sessions on agentic AI, RAG architectures, and autonomous system design.

Projects

- **Multi-Agent Autonomous Orchestration Platform (A2A-based)**

- Designed a research-driven platform to compose, coordinate, and persist autonomous multi-agent systems.
- Implemented agent discovery via agent cards, dynamic role assignment, and a supervisor agent to manage task execution and decision flow.
- Enabled full architecture reproducibility through YAML-based export/import of agent graphs.
- Integrated session-aware agents using Google ADK for long-running, stateful interactions.
- Focus Areas: Multi-agent coordination, workflow control, agent reproducibility
- Tech: LangChain, Google ADK, A2A SDK, FastAPI, PostgreSQL, React

- **Deep Research Agent with Sub-Agent Middleware**

- Built an autonomous research agent system capable of decomposing objectives into parallel research tasks.
- Used SubAgentMiddleware to spawn specialized research agents with tool access and memory.
- Integrated automated web exploration using DuckDuckGo search and Playwright, combined with structured progress tracking.
- Designed file and task management middleware for iterative knowledge accumulation.
- Focus Areas: Autonomous research, task planning, agent evaluation
- Tech: LangChain, LangGraph, DeepAgents, FastAPI, PostgreSQL

Education

Course / Degree	School / University	Grade / Score	Year
BTech with Honors	R.V.R & J.C College of Engineering	8.65	2024
Intermediate	Abhyaas Junior College	9.28	2020
10th	Bhashyam High School	9.5	2018

Internships

- DRDO

19-04-2024 to 19-01-2024

Research Intern
Gained hands-on experience in various software programs, increasing proficiency and expanding technical skill set.
Developed an desktop application using Qt to observe and analyze communication over 1553 protocol.

Technical Skills

- Programming**
Python, SQL, JavaScript, Java, C
- AI / ML & Research**
PyTorch, TensorFlow, Deep Learning, Computer Vision, Scikit-learn, NumPy, Pandas
- Autonomous Agents & LLMs**
LangChain, LangGraph, Google ADK, RAG, Agentic AI, Multi-Agent Systems, Hugging Face, Transformers, vLLM
- Automation & Integration**
API-based automation, Tool-using agents, Workflow orchestration
- Backend & Systems**
FastAPI, SQLAlchemy, REST APIs, Authentication & RBAC
- Databases**
PostgreSQL, MongoDB, SQLite, MySQL
- Cloud & DevOps**
GCP (Cloud Run, Cloud SQL, VPC), Docker, Git, CI/CD

Soft Skills

- Research-oriented problem solving
- Experimental rigor with practical execution
- Cross-functional collaboration
- Clear technical communication
- Rapid learning and framework adaptation

Achievements & Awards

- Received “On-the-Spot Team Award” for exceptional performance and collaboration.
- Secured 2nd Place in TCS Internal AI Hackathon, competing against cross-team AI solutions.

Certifications

- Cloud and DevOps from EPAM
- Project management from Google
- AWS Academy Cloud Foundations
- Java Full Stack from Wipro
- Advanced Data Analytics from Google